Chapter 4

Thematic Relevance Scoring

The first empirical objective of this study is to transform the textual content of news articles into quantitative measures of their relevance to a predefined set of financial risk categories. The analysis focuses on five candidate themes: cybersecurity risk, climate and environmental risk, pandemic risk, supply-chain risk, and artificial-intelligence-related risk.
For each article (i), theme (k), and scoring methodology (m), a continuous relevance score is defined as

$$S_{i,k}^{m} \in [0, 1], \tag{4.1}$$

where higher values indicate stronger evidence that article (i) is related to risk category (k).

This distinction is important: each article is not necessarily assigned to a single topic as it may simultaneously contain information about multiple risk categories, and the intensity of its relevance may vary considerably. The thematic-scoring problem is therefore intrinsically multi-label and continuous.

Moreover, the resulting scores depends on the methodology used. A keyword-based score measures lexical evidence, a weakly supervised score measures the intensity of rule-based contextual evidence, and the supervised model produces a continuous score derived from the logistic classifier.
The use of multiple methodologies serves two purposes. First, it provides alternative measurements of the same latent construct, namely thematic relevance. Second, it allows the strengths and limitations of increasingly sophisticated text-analysis approaches to be assessed empirically.
The three final methodologies are:

1. keyword-based scoring;
2. weakly-supervised scoring;
3. supervised scoring.

The progression from the first to the third method represents a gradual transition from explicit lexical rules to contextual rules and, finally, to patterns learned statistically from human-labelled observations.

4.1 Thematic Framework

The five candidate risk categories were defined ex ante and stored in the project database together with an initial set of seed terms. The categories are:

1. Cybersecurity Risk;
2. Climate / Environmental Risk;
3. Pandemic Risk;
4. Supply Chain Risk;
5. AI-Related Risk.

Seed terms include expressions such as *cybersecurity*, *data breach*, and *ransomware* for cyber risk; *climate change*, *carbon*, and *emissions* for climate risk; *pandemic*, *COVID*, and *epidemic* for pandemic risk; *supply chain*, *shortage*, and *logistics* for supply-chain risk; and *artificial intelligence*, *machine learning*, *algorithmic*, and *generative AI* for AI-related risk.
These seed terms provide an initial operational definition of the categories, but they are not treated as the final thematic classification. In particular, lexical occurrence does not necessarily imply substantive risk relevance. This limitation motivates the progressive development of the three scoring approaches described below.

4.2 Keyword-Based Scoring

The keyword method's purpose is to quantify direct lexical evidence connecting an article to a candidate risk category.
If an article contains expressions associated with a theme, its score increases with the strength of the lexical match in the full text. If no thematic expression appears, the score is zero.
Therefore, the keyword method is treated as a baseline since lexical matching cannot reliably distinguish substantive risk discussion from incidental or ambiguous uses of the same expressions.
For example, an environmental term may occur in a context unrelated to financial climate risk, while an article may discuss a risk substantively without using the exact expressions contained in the initial seed list.

4.2.1 Keyword Score

The implementation relies on PostgreSQL full-text search. Single-word seed terms are converted using `plainto_tsquery`, whereas multi-word expressions are represented using `phraseto_tsquery`. The resulting expressions are combined through an OR query and matched against the article body.
For an article (i) and theme (k), the raw lexical relevance measure is obtained through PostgreSQL's `ts_rank`. Articles with no thematic match receive

$$S_{i,k}^{KW} = 0. \tag{4.2}$$

For matching articles, the raw rank is normalized using the 99th percentile of the rank distribution among matching observations for the corresponding theme. The final score is

$$S_{i,k}^{KW} = \min\left(\frac{R_{i,k}}{P99_k}, 1\right) \tag{4.3}$$

where $R_{i,k}$ is the raw `ts_rank` value and $P99_k$ is the 99th percentile of positive ranks for theme k. The use of the 99th percentile rather than the maximum makes the normalization less sensitive to extreme observations and values exceeding this reference point are clipped at one.

4.2.2 Limitations

Manual inspection revealed an important limitation of direct lexical matching: the same expression may have substantially different meanings depending on context.
The keyword score therefore measures *lexical thematic evidence*, not a complete semantic assessment of whether an article discusses the corresponding financial risk.
This motivated the introduction of contextual rules through weak supervision.

4.3 Weakly-Supervised Scoring

This method introduced labeling functions that evaluate different forms of textual evidence, including thematic expressions, explicit risk terminology, title evidence, local proximity, and contextual patterns. The key idea is that the relevance of an article to a given risk category depends also on how specific thematic terms are and on the context in which they appear.

The approach is called *weak supervision*. Thematic signals are generated automatically through predefined, imperfect rules and were developed and refined using human inspected observations during the development stage.

4.3.1 Contextual Labeling Functions

The labeling functions were built separately for each of the five risk categories. Each risk has its own vocabulary and, more importantly, its own sources of ambiguity.

For cyber risk, some terms were specific enough to work as strong signals. “Ransomware” and “data breach” are good examples. The same applied to terms that clearly referred to hacking, cyberattacks, security breaches, or cyber threats. Other words were less reliable when they appeared alone. In those cases, the rules also looked at the title and at the words surrounding the cyber term. A cyber-related expression appearing close to words about attacks, vulnerabilities, threats, losses, or damage was more likely to indicate an actual cyber risk. This helped separate articles about real cyber incidents from articles that only mentioned security in a broader sense.

Climate risk was more difficult because environmental language appears in many articles that are not really about climate risk. Terms such as “climate risk” and “climate crisis” were clear signals. The rules also covered physical and transition risks. Physical risk included cases where climate events were linked to damage or economic effects. Transition risk covered topics such as climate policy, regulation, and the energy transition. Generic environmental words were treated more carefully. On their own, they were not enough. They had to appear close to terms related to damage, disruption, exposure, risk, or economic effects.

For pandemic risk, direct references to pandemics and infectious diseases were the starting point. The rules also looked at what happened because of the health event. This included restrictions, public health effects, economic losses, and disruptions. This mattered because words related to epidemics can sometimes appear in a completely different context or be used as a metaphor. A specific pandemic term could give a strong signal on its own. Less specific terms needed context showing that the article was actually discussing a health event and its effects.

Supply-chain risk had a similar issue with common business vocabulary. Words such as "shipping," "suppliers," "logistics," or "shortages" appear regularly in business news, even when there is no real supply-chain problem. They were not enough by themselves. More specific expressions, such as "supply chain disruption," were stronger signals. The rules also covered shortages, transport problems, logistics constraints, supplier issues, production bottlenecks, capacity problems, and restrictions affecting the movement or availability of goods. Here, the words around the term made a big difference. A reference to a supplier, for example, meant much more when it appeared next to language describing delays, shortages, or production problems.

AI risk needed stricter rules. An article can talk extensively about artificial intelligence without saying anything about AI-related risk. It may simply describe a new technology, a company, an AI product, or a new application. For this reason, the presence of AI terminology alone was not enough. Expressions such as "AI risk," "AI safety," and "algorithmic bias" were treated as strong signals. More general references to AI needed to appear close to language about negative effects, including threats, misinformation, privacy, cybersecurity, copyright, bias, safety, or job displacement. There was also another problem with AI articles. AI can be the source of a risk, but it can also be used to solve one. These cases should not be treated in the same way. An article about using AI to detect cyberattacks, predict climate events, or deal with supply-chain problems is not necessarily about AI risk. The rules were designed to capture this difference instead of counting every article where AI and risk-related words happened to appear together.

The same basic idea was used for all five categories. Very specific terms could provide enough evidence on their own. Broader or more ambiguous terms needed support from the surrounding text. We introduced this distinction after looking at the mistakes made by earlier versions of the labeling system. Many of the false positives came from keywords that were technically related to a risk category but were being used in a different context.

The frozen Weak Supervision V3 rules came out of this process. They were not just a longer version of the original keyword lists. The rules also considered how specific a term was, what appeared around it, and whether related concepts occurred close enough to suggest that they were actually connected in the article.

4.3.2 Iterative Development and Error Analysis

The weak-supervision system was developed iteratively. The general workflow was

$$\text{initial rules} \rightarrow \text{manual validation} \rightarrow \text{error analysis} \rightarrow \text{rule revision} \rightarrow \text{re-evaluation}. \tag{4.4}$$

The rules were refined step by step by looking at the false positives and false negatives produced by each version. When the same type of error appeared more than once, the rule behind it was changed rather than the individual article. The aim was to make each change useful for new articles as well, not only for the examples already examined during the error analysis. After several rounds of testing and adjustment, the final weak-supervision rules were frozen and used to build the continuous relevance score.

4.3.3 Weak-Supervision Score

The frozen Weak Supervision V3 rules provide the qualitative structure for the continuous weakly supervised score. The final continuous specification combines three quantities derived from the contextual evidence identified in an article:

- the frequency of strong thematic evidence;
- the density of strong thematic evidence;
- the proximity between relevant thematic and risk expressions.

Let the strong-term frequency be denoted by f, the strong-term density per 1,000 text units by d_s, and the minimum relevant proximity distance by d. Three normalized components are constructed:

$$F = 1 - e^{-f}, \tag{4.5}$$

so that additional occurrences increase the score while their marginal contribution progressively decreases,

$$D = \frac{d_s}{1 + d_s}, \tag{4.6}$$

which maps the density of strong evidence to the interval [0,1) and

$$P = \max\left(0, 1 - \frac{d}{50}\right) \tag{4.7}$$

Under this specification, closer thematic and risk expressions generate stronger proximity evidence, while pairs separated by 50 or more text units receive no contribution from the proximity component. The proximity window of 50 is part of the frozen Weak Supervision V3 configuration selected during model development.

The three components are combined differently across themes, consistently with the theme-specific structure of the contextual rules. For cybersecurity, climate, and supply-chain risk, the continuous score is

$$S_{i,k}^{WS} = 1 - (1 - F)(1 - P)(1 - 0.35D), \tag{4.8}$$

For pandemic risk, the frozen specification combines frequency and density without the proximity component:

$$S^{WS}_{i,\text{Pandemic}} = 1 - (1 - F)(1 - 0.35D). \tag{4.9}$$

For AI-related risk, the continuous score is based on the proximity component:

$$S^{WS}_{i,\text{AI}} = P. \tag{4.10}$$

This more restrictive specification is consistent with the particular ambiguity of AI terminology. Frequent references to artificial intelligence do not necessarily constitute evidence of AI-related risk; what matters is whether AI-related expressions occur in the relevant adverse-risk context.

The coefficient 0.35 represents the density-reinforcement parameter of the continuous specification selected during development. Together with the proximity window of 50, it was fixed as part of the final configuration before evaluation on the independent final test set.

The resulting weakly supervised score lies in the interval [0,1]. It should not be interpreted as a calibrated probability that an article belongs to a given risk category. Instead, it represents a continuous rule-based measure of the strength of thematic evidence identified by the frozen Weak Supervision V3 specification.

4.4 Supervised Scoring

The supervised approach differs conceptually from the previous two methods.
Keyword scoring relies on explicit lexical matching. Weak supervision relies on contextual rules specified during model development. The supervised classifier instead learns statistical regularities from manually labelled article–theme observations.
Each training observation is an article–theme pair *(i,k)* associated with a human label

$$y_{i,k} \in 0, 1. \tag{4.11}$$

The objective is to learn which textual patterns distinguish substantively relevant article–theme pairs from irrelevant ones.

4.4.1 Text Representation

The final supervised architecture uses a `FeatureUnion` combining three complementary text representations.

First is a shared word-level TF–IDF representation using word unigrams and bigrams, English stop-word removal, sublinear term frequency, a minimum document frequency of two, a maximum document-frequency threshold of 0.95, and a maximum of 25,000 features.

Second is a shared character-level TF–IDF representation using $char_wb$ character 3- to 5-grams, a minimum document frequency of two, sublinear term frequency, and a maximum of 25,000 features.

Third, explicit theme–text interactions are introduced using unigram and bigram features generated through a custom analyzer generated through a custom analyzer, with a minimum document frequency of two, a maximum document frequency of 0.95, a maximum of 50,000 features, sublinear term frequency, and L_2 normalization.

The theme–text component is important because the same article is evaluated with respect to multiple candidate risks. A textual expression may therefore provide strong evidence for one theme while being irrelevant to another.

4.4.2 Logistic Regression

The combined TF–IDF representation is passed to a logistic-regression classifier.
For a feature vector (x), the model computes

$$z = \beta_0 + \sum_{j=1}^{p} \beta_j x_j, \tag{4.12}$$

followed by the logistic transformation

$$p(y = 1|x) = \frac{1}{1 + e^{-z}}. \tag{4.13}$$

The frozen model uses:

- $C = 4$;
- L_2 regularization;
- the `liblinear` solver;
- balanced class weights;
- a maximum of 2,000 iterations;
- random seed 42.

Balanced class weights were used because positive thematic observations were substantially less frequent than negative observations.
The frozen classification threshold is

$$\tau = 0.34. \tag{4.14}$$

Thus, for binary evaluation,

$$\hat{y}_{i,k} = \mathbb{I}\left[S_{i,k}^{SUP} \geq 0.34\right]. \tag{4.15}$$

For the full corpus, the model keeps the continuous `predict_proba` score instead of using only the final 0/1 classification. This score shows how strongly the model associates an article with a given risk

theme. Since no additional probability calibration is applied, it is used as a continuous model score rather than as an estimate of the article's true probability of belonging to that risk category.

4.5 Full-Corpus Production

The final analytical corpus contains

$$N = 2,249,255 \tag{4.16}$$

valid and non-duplicate articles.
Each article is evaluated against all five candidate themes. Therefore, each methodology produces

$$2,249,255 \times 5 = 11,246,275 \tag{4.17}$$

article–theme scores.
With three frozen methodologies, the complete final scoring view contains

$$11,246,275 \times 3 = 33,738,825 \tag{4.18}$$

method–article–theme observations.

These scores answer the question of *which risk an article is related to and how strongly*. They do not, however, provide a general representation of the semantic content of the article.
The next chapter therefore introduces document embeddings, which map each article into a continuous semantic vector space independently of the five predefined risk categories.